

Rung One of the Word-Complexity Ladder: a Machine-Verified Exclusion of an Automatic Collatz Itinerary by Mahler’s Method in Degree One

M. Sharpe*

August 28, 2026

Abstract

Let $w = 110110011011110011\dots$ be the fixed point of the substitution $\sigma: 1 \mapsto 110, 0 \mapsto 011$. Its density of ones is $2/3 > \log 2 / \log 3$, so it lies inside the band where a divergent orbit of the Terras map T is allowed to live. We prove that no positive integer has Terras parity itinerary w , or any shift of w . Equivalently, the 2-adic Mahler value $F_2(8/9)$, where $F(z) = \sum_i w_i z^i$ satisfies $F(z) = (z + z^2)/(1 - z^3) + (1 - z^2)F(z^3)$, is irrational. The proof is Mahler’s method with a degree-one auxiliary form, carried out entirely in the language of integer divisibility: a Padé pair $(p_0, p_1) = (1 + z^3 - z^4, -1 + z - z^2 + z^3)$ found on the first ten letters of w , the functional equation iterated along $\alpha_k = (8/9)^{3^k}$, and a height ledger that closes because $2^{27} > 28 \cdot 9^7$. Every step is machine-verified in Lean 4 (`Collatz/Rung1.lean`, axiom-free). The earlier reduction paper recorded the truncation approach to $F_2(8/9)$ as provably insufficient (irrationality exponent $0.946 < 1$) and the 2-adic transposition of Mahler’s method as the open work item; we show the transposition is short — in the ultrametric setting the analytic lemma is trivial and non-vanishing is free, so neither a zero estimate nor the transcendence of F over $\mathbb{Q}(z)$ is needed. This is the first exclusion of a genuinely aperiodic itinerary class for the $3x + 1$ problem, and, to our knowledge, the first formalization of a Mahler-method argument in a proof assistant. We are explicit about what is classical: the irrationality of p -adic Mahler values at rational points inside the unit disc is a known type of result. Finally we begin rung two: an elementary prefix-power criterion (the Diophantine content of the parity bijection, machine-verified) excludes every Sturmian itinerary whose slope has partial quotients ≥ 2 infinitely often, for slopes up to 0.789 — conditional on two classical facts, Lagrange’s best-approximation property and the three-distance theorem, which are isolated as hypotheses of a machine-verified level theorem.

1 Introduction

The word-complexity ladder [3] orders potential divergent orbits of the Terras map

$$T(n) = \begin{cases} n/2 & n \text{ even,} \\ (3n + 1)/2 & n \text{ odd} \end{cases}$$

*msharpe@irendi.com. The mathematics, the Lean formalization, and this text were produced in collaboration with Claude (Anthropic). The Collatz conjecture remains open, and nothing in this paper claims otherwise.

by the complexity of their parity itineraries $w_t = T^t(n) \bmod 2$. Rung zero — a divergent orbit’s itinerary is never eventually periodic — is formal. The next rung consists of *automatic* itineraries, fixed points of constant-length substitutions, and [3] reduced the running example exactly to one arithmetic statement: for $w = \sigma^\infty(1)$ with $\sigma(1) = 110$, $\sigma(0) = 011$,

$$\text{some integer has itinerary } w \iff F_2(8/9) \in \mathbb{Q}, \quad (1)$$

where $F(z) = \sum_{i \geq 0} w_i z^i \in \mathbb{Z}[[z]]$ and F_2 is evaluation in $\mathbb{Q}_2$ (convergent since $|8/9|_2 = 1/8$). The reduction paper also proved that the obvious approach fails: the partial sums of $F_2(8/9)$ have denominators 9^N and 2-adic error 2^{-3N} , an irrationality exponent of $(3 \log 2)/(2 \log 3) = 0.946 < 1$, so no Liouville-type argument from truncation can decide (1). It identified Mahler’s method as the route and its 2-adic transposition as “bounded, classical work, but real work.”

This paper does that work, and finds it shorter than expected.

Theorem 1.1 (Rung one, running example). *For every $n \in \mathbb{N}$ and every $s \in \mathbb{N}$, the Terras itinerary of n is not the s -shift of w : it is false that $T^t(n) \bmod 2 = w_{s+t}$ for all t . In particular no Collatz orbit has itinerary w , and no orbit ever synchronises with w .*

Theorem 1.2 (Irrationality). $F_2(8/9) \notin \mathbb{Q}$.

Both are formalized: `no_shifted_sigWord_itinerary` and `core` in `lean/Collatz/Rung1.lean`, depending only on `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`.

What is new, and what is not. We want to be precise.

- *New as a Collatz statement.* Theorem 1.1 is the first exclusion of an aperiodic itinerary class: rung zero excluded periodic words, and Theorem 1.1 excludes an automatic word of supercritical density together with all its tails. The density corridor [8] permits such an orbit; the arithmetic of F forbids it.
- *New as a formalization.* To our knowledge no Mahler-method argument (transcendence or irrationality of values of Mahler functions) has previously been checked by a proof assistant. The formalization is possible because the whole argument is stated with integers and divisibility by powers of 2; no p -adic analysis and no Mathlib theory of $\mathbb{Z}_2$ is used.
- *Not new as number theory.* That values of Mahler functions at rational (or algebraic) points inside the p -adic unit disc are irrational or transcendental is a classical theme: Mahler [5], Loxton and van der Poorten [4], Bundschuh and Väänänen [1] and others. Our Section 6 places the proof in that context. What we add is the observation, of some independent interest, that for *irrationality* at a point with $|\alpha|_p < 1$ the method needs only a degree-one auxiliary form, and that the two ingredients which make the archimedean method delicate — the analytic lemma and the non-vanishing of the auxiliary function along the orbit — are both trivial ultrametrically.
- *The real content of the 0.946 barrier.* The truncation route trades height 9^L for 2-adic gain 2^{3L} . A degree-one Padé form trades height $9^{4 \cdot 3^k}$ (plus a sunk cost $9^{(5/2) \cdot 3^k}$ independent of the degree) for gain $2^{9 \cdot 3^{k+1}}$: per unit of degree the ratio is $6 \log 2 / \log 9 = 1.89 > 1$. This is the classical “squaring” that Mahler’s iteration buys; the barrier was a fact about truncation, not about the number.

Organisation. Section 2 fixes notation and proves the exact finite form of the functional equation and of the block structure of the correction term. Section 3 sets up the tower of forced rationals. Section 4 gives the certificate and the exact 2-adic size. Section 5 does the height accounting and proves the theorems. Section 6 discusses the relation to Mahler’s method and the extent of generality. Section 7 records the next rung. Throughout, lemma names in `typewriter` refer to the Lean file.

2 Setup: the word, prefix sums, and the exact identities

2.1 The word

Define $w: \mathbb{N} \rightarrow \{0, 1\}$ by $w_0 = 1$ and, for $q \geq 0$,

$$w_{3q} = w_q, \quad w_{3q+1} = 1, \quad w_{3q+2} = 1 - w_q \quad (2)$$

(`sigWord`, `sigWord_three`, `sigWord_three_one`, `sigWord_three_two`). This is the fixed point $\sigma^\infty(1)$: the block $w_{3q}w_{3q+1}w_{3q+2}$ is $\sigma(w_q)$. The first ten letters are 1, 1, 0, 1, 1, 0, 0, 1, 1, 1 (`sigWord_first_ten`); every block has exactly two ones.

2.2 Homogenised prefix sums

For $a, b \in \mathbb{Z}$ and $L \in \mathbb{N}$ put

$$S_L(a, b) = \sum_{q < L} w_q a^q b^{L-1-q}, \quad G_L(a, b) = \sum_{q < L} a^q b^{L-1-q},$$

so that $S_L(a, b) = b^{L-1} P_L(a/b)$ where $P_L(z) = \sum_{q < L} w_q z^q$ is the degree- $< L$ prefix of F , and G_L is the same for the all-ones word. Both are defined by the back-recursions $S_0 = 0$, $S_{L+1} = b S_L + w_L a^L$ and $G_0 = 0$, $G_{L+1} = b G_L + a^L$ (`preS`, `geomS`). The geometric identity

$$(b - a) G_L(a, b) = b^L - a^L \quad (3)$$

is immediate by induction (`geomS_mul`).

Lemma 2.1 (Functional equation, finite form; `preS_three`). *For all $a, b \in \mathbb{Z}$ and $L \in \mathbb{N}$,*

$$S_{3L}(a, b) = (ab + a^2) G_L(a^3, b^3) + (b^2 - a^2) S_L(a^3, b^3).$$

Proof. Induction on L . Unrolling the recursion three times, $S_{3L+3} = b^3 S_{3L} + w_{3L} a^{3L} b^2 + w_{3L+1} a^{3L+1} b + w_{3L+2} a^{3L+2}$, and by (2) the three new terms are $a^{3L}(w_L b^2 + ab + (1 - w_L) a^2) = a^{3L}(ab + a^2 + w_L(b^2 - a^2))$. On the right, $G_{L+1}(a^3, b^3) = b^3 G_L + a^{3L}$ and $S_{L+1}(a^3, b^3) = b^3 S_L + w_L a^{3L}$, and the two sides agree by the induction hypothesis and a ring identity. $\square$

Dividing by b^{3L-1} and letting $L \rightarrow \infty$ formally, this is $F(z) = (z+z^2)/(1-z^3) + (1-z^2)F(z^3)$ at $z = a/b$; but we never pass to the limit. Every statement below is about finite L .

2.3 The correction term of a word, and the block formula

For an arbitrary word $u: \mathbb{N} \rightarrow \{0, 1\}$ define

$$d_u(0) = 0, \quad d_u(T+1) = 3^{u_T} d_u(T) + 2^T u_T, \quad j_u(T) = \sum_{t < T} u_t$$

(wordD, wordJ). If the itinerary of n agrees with u on $t < T$, then $d_u(T)$ is the correction term $d(T, n)$ of the exact orbit identity $2^T T^T(n) = 3^j n + d$ of [8], and $j_u(T)$ is its odd-step count j (dcoef_eq_wordD; this is the cocycle law of [8] at $T = 1$). The word-level cocycle law is

$$d_u(s+T) = 3^{j_{u'}(T)} d_u(s) + 2^s d_{u'}(T), \quad j_u(s+T) = j_u(s) + j_{u'}(T), \quad u' = u(s + \cdot) \quad (4)$$

(wordD_add).

Lemma 2.2 (Block formula; wordD_sigWord_three). *For all L , $j_w(3L) = 2L$ and*

$$d_w(3L) = 10(9^L - 8^L) - 5S_L(8, 9).$$

Proof. Induction on L : appending the block $\sigma(w_L)$ to a prefix of length $3L$ with correction d gives $9d + 5 \cdot 8^L$ if $w_L = 1$ (append 1, 1, 0) and $9d + 10 \cdot 8^L$ if $w_L = 0$ (append 0, 1, 1), i.e. $d_w(3L+3) = 9d_w(3L) + (10 - 5w_L)8^L$, which matches the recursion $S_{L+1}(8, 9) = 9S_L(8, 9) + w_L 8^L$. $\square$

2.4 From an itinerary to a congruence for every prefix

Proposition 2.3 (The link; link). *Suppose the itinerary of n is the s -shift of w . Put*

$$q = 3^{j_w(s)}, \quad p = 2^s n - d_w(s), \quad D_0 = 5q, \quad u_0 = 9(p + 10q).$$

Then for every L with $3L \geq s$,

$$2^{3L} \mid 9^L(p + 10q) - 5q S_L(8, 9). \quad (5)$$

Proof. Write $3L = s + T$. By (4) with $u = w$, the identification $d_{w'}(T) = d(T, n)$, $j_{w'}(T) = j$ for the orbit of n , and the exact orbit identity $2^T T^T(n) = 3^j n + d(T, n)$,

$$3^{j_w(s)} d_w(3L) = 3^{j_w(s)+j} (d_w(s) - 2^s n) + 2^{s+T} 3^{j_w(s)} T^T(n),$$

and $j_w(s) + j = j_w(3L) = 2L$. Substituting the block formula for $d_w(3L)$ and using $2^{3L} \mid 10 \cdot 8^L q$ gives (5). $\square$

In 2-adic language, (5) for all L says exactly that $F_2(8/9) = u_0/(9D_0) \cdot 9 = (p+10q)/q \cdot 9/5$, i.e. that $x(w) = p/q$ is the rational with odd denominator $3^{j_w(s)}$ obtained by pulling n back s steps through the 2-adic Terras map. Shifts therefore cost nothing: they only change the odd denominator. For $s = 0$, $q = 1$ and $p = n$.

3 The tower

Put $a_k = 8^{3^k} = 2^{3^{k+1}}$ and $b_k = 9^{3^k}$ (1vA, 1vB), so $\alpha_k = a_k/b_k = (8/9)^{3^k}$, $a_{k+1} = a_k^3$, $b_{k+1} = b_k^3$, $|\alpha_k|_2 = 2^{-3^{k+1}}$. Given $D_0, u_0 \in \mathbb{Z}$ define (1vD, 1vU)

$$\begin{aligned} D_{k+1} &= D_k (b_k^2 - a_k^2)(b_k^3 - a_k^3), \\ u_{k+1} &= b_k^2 \left((b_k^3 - a_k^3) u_k - D_k a_k (a_k + b_k) b_k \right). \end{aligned}$$

These are the numerator and denominator obtained by running the functional equation backwards: if $F_2(\alpha_k) = u_k/D_k$ then $F_2(\alpha_{k+1}) = (F_2(\alpha_k) - A(\alpha_k))/B(\alpha_k) = u_{k+1}/D_{k+1}$ with $A(z) = (z + z^2)/(1 - z^3)$, $B(z) = 1 - z^2$. We never use this interpretation; the finite statement is the following.

Definition 3.1 (Hyp). $\text{Hyp}_k(L_0)$ is the statement: for all $L \geq L_0$,

$$2^{3^{k+1}(L+1)} \mid u_k b_k^L - D_k S_{L+1}(a_k, b_k).$$

For $k = 0$ and $D_0 = 5q$, $u_0 = 9(p+10q)$ this is (5) with $L+1$ in place of L , so Proposition 2.3 gives $\text{Hyp}_0(s)$ (the first few L are discarded harmlessly; only large L matter).

Proposition 3.2 (Transport; Hyp_succ). $\text{Hyp}_k(L_0) \Rightarrow \text{Hyp}_{k+1}(L_0)$.

Proof. Fix $L \geq L_0$ and apply Hyp_k at prefix length $3(L+1)$: $u_k b_k^{3L+2} - D_k S_{3(L+1)}(a_k, b_k) = 2^{3^{k+1} \cdot 3(L+1)} c$ for some $c \in \mathbb{Z}$. Write $a = a_k$, $b = b_k$. Multiplying by $b^3 - a^3$ and using Lemma 2.1 and (3) (with (a^3, b^3) and $L+1$),

$$(b^3 - a^3)(u_k b_k^{3L+2} - D_k S_{3(L+1)}(a, b)) = u_{k+1} b_{k+1}^L - D_{k+1} S_{L+1}(a_{k+1}, b_{k+1}) + D_k (ab + a^2) a^{3(L+1)}.$$

Both the left side and $a^{3(L+1)} = 2^{3^{k+2}(L+1)}$ are divisible by $2^{3^{k+2}(L+1)}$, hence so is the middle term. $\square$

Note that $b^3 - a^3$ is odd, so nothing is lost by multiplying through: the divisibility is of the product, and that is all we need downstream. By induction, $\text{Hyp}_0(L_0)$ implies $\text{Hyp}_k(L_0)$ for all k (Hyp_all).

4 The Padé certificate and the exact 2-adic size

Let $p_0(z) = 1 + z^3 - z^4$ and $p_1(z) = -1 + z - z^2 + z^3$, and their degree-4 homogenisations

$$P_0(a, b) = b^4 + a^3 b - a^4, \quad P_1(a, b) = -b^4 + ab^3 - a^2 b^2 + a^3 b$$

(P0, P1). The pair was found by solving the 9 homogeneous linear conditions “coefficients of $z^0, \dots, z^8$ in $p_0 + p_1 F$ vanish” in the 10 unknown coefficients; the solution is unique up to scale and has integer entries.

Lemma 4.1 (The certificate; pade_base). *As an identity in $\mathbb{Z}[a, b]$,*

$$b^9 P_0(a, b) + P_1(a, b) S_{10}(a, b) = a^9 (-b^4 + a(b^3 + a^2 b)).$$

Proof. Expand S_{10} with the first ten letters of w and compare coefficients; this is a finite computation, done in Lean by `simp` and `ring`. $\square$

In words: $E(z) = p_0(z) + p_1(z)F(z) = -z^9 + z^{10} + z^{12} + \dots$ has order exactly 9 at $z = 0$ with leading coefficient -1 . The lemma is the homogeneous form of this restricted to the prefix of length 10; longer prefixes only add terms divisible by a^{10} :

Lemma 4.2 (`pade_gen`). *For every $M \geq 0$ there is $R \in \mathbb{Z}[a, b]$ with*

$$b^{9+M}P_0(a, b) + P_1(a, b)S_{10+M}(a, b) = a^9(-b^{4+M} + aR).$$

Proof. Induction on M from Lemma 4.1, using $S_{L+1} = bS_L + w_L a^L$: the new term $P_1 w_L a^L$ with $L = 10 + M \geq 10$ is divisible by a^{10} . $\square$

Lemma 4.3 (Exact 2-adic size; `level_dvd`). *Let $v \geq 1$, $a = 2^v$, b odd, D odd, $M \geq 0$, and suppose*

$$2^{v(10+M)} \mid ub^{9+M} - DS_{10+M}(a, b).$$

Then $Z := P_0(a, b)D + P_1(a, b)u$ is a nonzero integer divisible by 2^{9v} .

Proof. Write $ub^{9+M} - DS_{10+M} = 2^{v(10+M)}t$ and take R from Lemma 4.2. Then

$$b^{9+M}Z = D(b^{9+M}P_0 + P_1S_{10+M}) + P_1(ub^{9+M} - DS_{10+M}) = 2^{9v}O, \quad O = D(-b^{4+M} + 2^vR) + P_1 2^{v(1+M)}t.$$

O is odd: $D b^{4+M}$ is odd and the other two terms are even. Hence $Z \neq 0$ (else $O = 0$), and $2^{9v} \mid b^{9+M}Z$ with b^{9+M} odd gives $2^{9v} \mid Z$. $\square$

This lemma is the whole of “Mahler’s analytic lemma” and “the zero estimate” in our setting. Archimedeanly one bounds $|E(\alpha_k)|$ from above by a Schwarz-lemma argument and must separately show $E(\alpha_k) \neq 0$ for infinitely many k , which is where the transcendence of F over $\mathbb{Q}(z)$ enters. Ultrametrically the size of $E(\alpha_k)$ is *exactly* $|\alpha_k|_2^9$ as soon as $|\alpha_k|_2 < |c|_2 = 1$ where $c = -1$ is the leading coefficient, because all higher coefficients are integers. Nothing else is needed.

5 The height ledger and the proofs

Lemma 5.1 (Size of Z ; `abs_Z_le`). *If $0 \leq a \leq b$ and $|D|, |u| \leq m$ then $|P_0(a, b)D + P_1(a, b)u| \leq 7b^4m$.*

Proof. $|P_0| \leq 3b^4$ and $|P_1| \leq 4b^4$ termwise. $\square$

Lemma 5.2 (Tower heights; `lv_bounds`). *Let $m_0 = \max(|D_0|, |u_0|)$ and $E_k = 3^k 9^{3 \cdot 3^k} m_0$. Then $|D_k|, |u_k| \leq E_k$ for all k .*

Proof. $E_0 = 729 m_0 \geq m_0$. With $0 \leq a_k \leq b_k$: $|D_{k+1}| \leq E_k b_k^5$ and $|u_{k+1}| \leq b_k^2(b_k^3 E_k + E_k \cdot 2b_k^3) = 3b_k^5 E_k$, while $E_{k+1} = 3b_k^6 E_k$ (`lvE_succ`). $\square$

Also D_k is odd for all k if D_0 is (`lvD_odd`): b_k is odd and a_k even.

Lemma 5.3 (The numeric inequality; `final_ineq`). *If $0 \leq m_0 \leq k$ then $7b_k^4 E_k < 2^{9 \cdot 3^{k+1}}$.*

Proof. The left side is $7 \cdot 3^k \cdot 9^{7 \cdot 3^k} m_0 \leq 7 \cdot 3^k \cdot 9^{7 \cdot 3^k} k$. Since $k + 1 \leq 3^k$ and $k < 9^k$,

$$7 \cdot 3^k k < 28 \cdot 3^k 9^k = 28 \cdot 27^k \leq 28^{k+1} \leq 28^{3^k},$$

so the left side is $< 28^{3^k} 9^{7 \cdot 3^k} = (28 \cdot 9^7)^{3^k} \leq (2^{27})^{3^k} = 2^{27 \cdot 3^k}$, using $28 \cdot 9^7 = 133\,923\,132 \leq 134\,217\,728 = 2^{27}$. $\square$

The margin is 0.22% per tower level, which is why the degree-4 pair (order 9) is used rather than the degree-2 pair (order 5): the ledger for general m is $3(2m + 1) \log 2$ against $(m + 5/2) \log 9$, positive for every $m \geq 2$, but the crude constants of Lemmas 5.1 and 5.2 eat the $m = 2$ margin.

Theorem 5.4 (core). *Let D_0 be odd, $u_0 \in \mathbb{Z}$, and $L_0 \in \mathbb{N}$. Then $\text{Hyp}_0(L_0)$ is false.*

Proof. Suppose $\text{Hyp}_0(L_0)$. Let $m_0 = \max(|D_0|, |u_0|)$ and $k = m_0$. By Proposition 3.2, $\text{Hyp}_k(L_0)$ holds; apply it at prefix length $10 + M$ where $9 + M = \max(L_0, 9)$. With $v = 3^{k+1}$, $a_k = 2^v$, b_k odd and D_k odd, Lemma 4.3 gives $2^{9 \cdot 3^{k+1}} \leq |Z_k|$ for $Z_k = P_0(a_k, b_k)D_k + P_1(a_k, b_k)u_k$. Lemmas 5.1 and 5.2 give $|Z_k| \leq 7b_k^4 E_k$, and Lemma 5.3 gives $7b_k^4 E_k < 2^{9 \cdot 3^{k+1}}$. Contradiction. $\square$

Proof of Theorem 1.1. If the itinerary of n were the s -shift of w , Proposition 2.3 would give $\text{Hyp}_0(s)$ with $D_0 = 5 \cdot 3^{j_w(s)}$ odd, contradicting Theorem 5.4. $\square$

Proof of Theorem 1.2. $F_2(8/9) \in \mathbb{Z}_2$, so if it were rational it would be u_0/D_0 with D_0 odd, and then $\text{Hyp}_0(0)$ would hold (the partial sums $S_{L+1}(8, 9)/9^L$ converge to $F_2(8/9)$ with error $O(2^{-3(L+1)})$). Theorem 5.4 forbids it. (The Lean file proves Theorem 5.4 as its central statement and derives Theorem 1.1; the two-line passage from $F_2(8/9) \in \mathbb{Q}$ to $\text{Hyp}_0(0)$ is not formalized, since F_2 itself is not defined in the file.) $\square$

6 Relation to Mahler's method; scope of the argument

The classical shape. Mahler's method [5, 6] proves transcendence of $F(\alpha)$ for F satisfying $F(z) = A(z) + B(z)F(z^d)$ by constructing an auxiliary polynomial $P(z, F(z))$ of degree m in F with high vanishing order $N \sim m^2$ at 0 (Siegel's lemma), evaluating along α^{d^k} , and comparing the analytic upper bound $|P(\alpha_k, F(\alpha_k))| \ll |\alpha_k|^N$ with a Liouville lower bound from the algebraic height, which grows like $d^k \cdot m$. The two delicate points are a zero estimate guaranteeing $P(\alpha_k, F(\alpha_k)) \neq 0$ for infinitely many k (this uses that F is transcendental over $\mathbb{Q}(z)$), and the Schwarz-type analytic lemma.

What changes 2-adically, and why degree one suffices. For irrationality one takes $m = 1$: $E = p_0 + p_1 F$ with $\deg p_i \leq m'$, $N = 2m' + 1$. Archimedeanly $m = 1$ is hopeless (it recovers only Liouville-grade approximations). Ultrametrically it is enough because the height grows like $d^k \cdot m'$ while the gain grows like $d^k \cdot N \cdot v_2(\alpha)$, and for $\alpha = 8/9$, $d = 3$: $3(2m' + 1) \log 2 > (m' + 5/2) \log 9$ for all $m' \geq 2$. The $5/2$ is the cost of transporting the rational value down the tower; it is independent of m' , which is exactly why the truncation approach (which pays the analogous cost at every step) sees exponent 0.946 while the Padé form sees 1.89 per unit degree. And, as Lemma 4.3 shows, both delicate points disappear: with integer coefficients and $|\alpha_k|_2 \rightarrow 0$, the value of E has *exactly* the 2-adic size of its leading term,

so non-vanishing is automatic and the transcendence of F over $\mathbb{Q}(z)$ is never used. (It is true — w is aperiodic — but the certificate $c = -1 \neq 0$ is the only fact about the leading behaviour of E that the proof consumes.)

Where this sits in the literature. Irrationality and transcendence of p -adic values of Mahler functions at points inside the unit disc, and irrationality measures for them, are classical: see Loxton–van der Poorten [4], Bundschuh and Väänänen [1], and the function-field development by Fernandes [2]. We have not located a statement that covers $F_2(8/9)$ for this specific F verbatim, but we would be surprised if the general form — $F \in \mathbb{Z}[[z]]$ Mahler of degree d over $\mathbb{Q}(z)$, not rational, $\alpha \in \mathbb{Q}$ with $0 < |\alpha|_p < 1$, $\Rightarrow F_p(\alpha) \notin \mathbb{Q}$ — were not known to experts, and we make no priority claim for it. What we claim is the Collatz consequence, the explicit finite certificate, and the formalization.

The whole rung. The argument uses only: (i) w is the fixed point of a constant-length substitution of length ℓ with every block containing the same number a of ones, which gives the block formula with $(2^\ell, 3^a)$ in place of $(8, 9)$ and a functional equation of degree ℓ with polynomial coefficients; (ii) $|2^\ell/3^a|_2 = 2^{-\ell} < 1$; (iii) a Padé certificate, found by linear algebra on the first $2m' + 2$ letters, with odd leading coefficient (or leading coefficient of known 2-adic size). The supercritical condition $2^\ell < 3^a$ is what makes the real value also converge, but it plays no role in the 2-adic proof: the argument excludes *subcritical* uniform-weight automatic words as itineraries too, which is consistent, since those are already excluded by the density corridor. A parametric Lean theorem with the certificate as an explicit input is the natural next formalization, and would turn rung one from one word into the class.

7 The next rung: Sturmian itineraries

Rung two consists of morphic and Sturmian itineraries. The Mahler route continues to non-uniform block weights via Hermite–Padé forms for vector Mahler systems, with the same trivial ultrametric analytic lemma. But for Sturmian words there is an elementary route with no functional equation at all, whose threshold lands exactly on the critical line; it is machine-verified up to two classical facts of Diophantine approximation, which we isolate as hypotheses.

7.1 The prefix-power criterion

Proposition 7.1 (`prefix_power`, `prefix_power_bound`). *Let $\ell \leq M$. If the itinerary of n is ℓ -periodic on its first M letters, i.e. $T^t(n) \equiv T^{t+\ell}(n) \pmod{2}$ for all $t + \ell < M$, then either $T^\ell(n) = n$, or*

$$2^{M-\ell} \leq \max(n, T^\ell(n)), \quad \text{hence} \quad 2^M \leq 2^\ell n + 3^o(n + 2^\ell),$$

where o is the number of odd steps among the first ℓ .

Proof. n and $T^\ell(n)$ have the same first $M - \ell$ parities, so they are congruent modulo $2^{M-\ell}$ by the parity bijection [8]; two distinct naturals congruent modulo $2^{M-\ell}$ differ by at least $2^{M-\ell}$. The second form uses the master inequality $2^\ell(T^\ell(n) + 1) \leq 3^o(n + 2^\ell)$. $\square$

In logarithmic form: if the itinerary begins with an e -th power (not necessarily integral) of a word of length ℓ and ones-density δ , then $(e - 1)\ell \leq \delta\ell \log_2 3 + \log_2(n + 1) + O(1)$. The threshold exponent $1 + \delta \log_2 3$ equals 2 precisely at $\delta = \log 2 / \log 3$. Below the critical line squares are impossible for large ℓ ; above it one needs $e > 2$. This is the Diophantine content of the parity bijection: the 2-adic realizations of periodic words are rationals $d(u)/(2^\ell - 3^o)$, and an integer cannot lie 2-adically within 2^{-M} of one unless it is as large as the approximation is good.

We note in passing what this criterion is *not*: for a rational (here, integer) target, Roth-, Ridout- and Subspace-theorem improvements of Liouville's inequality have no content, since the lower bound $|Nq - p| \geq 1$ is already exact. The lever is the word exponent, not the Diophantine exponent.

7.2 Sturmian words and their long periodic stretches

Fix $\alpha \in (0, 1)$ irrational and $\rho \in \mathbb{R}$, and let $w_n = \lfloor (n + 1)\alpha + \rho \rfloor - \lfloor n\alpha + \rho \rfloor \in \{0, 1\}$ be the mechanical word; every Sturmian word is of this form, its ones-density is α , and every shift of it is again mechanical with the same slope. Let p_k/q_k be the convergents of α , with partial quotients a_k , and $\delta_k = |q_k\alpha - p_k|$; the sign of $q_k\alpha - p_k$ alternates, and we call k *positive* when it is > 0 .

Fix a positive k , write $q = q_k$, $\delta = \delta_k$, $Q = q_{k+1}$, and call n a *visit* if $\{n\alpha + \rho\} \geq 1 - \delta$. Two elementary facts (`floor_shift`, `mech_shift`, `visit_sep`):

- (i) $\lfloor (n + q)\alpha + \rho \rfloor = \lfloor n\alpha + \rho \rfloor + p + [n \text{ is a visit}]$, hence $w_{n+q} = w_n$ unless n or $n + 1$ is a visit.
- (ii) If $\|m\alpha\| \geq \delta$ for all $0 < m < Q$, two visits are at least Q apart: their fractional parts lie in a half-open interval of length δ , so their difference $n' - n$ satisfies $\|(n' - n)\alpha\| < \delta$.

The hypothesis of (ii) is Lagrange's best-approximation property of convergents: q_{k+1} is the least $m > 0$ with $\|m\alpha\| < \|q_k\alpha\|$. The second classical input is the three-distance theorem [7]: among any $q_k + q_{k+1}$ consecutive points $\{n\alpha + \rho\}$, every gap is at most $\|q_k\alpha\|$, so every window of length $G = q_k + q_{k+1}$ contains a visit.

Consequently, if n_0 is the first visit, $n_0 < G$, and the word is q -periodic on the run $[n_0 + 1, n_0 + Q - 1]$: a run of length $Q - 2$ starting at $s = n_0 + 1 \leq G$.

7.3 The size of a tail

Suppose N has itinerary w with α above the critical line, i.e. $2 \leq 3^\alpha$. The exact orbit identity gives $2^s T^s(N) = 3^{j_s} N + d_s$ with $j_s = \lfloor s\alpha + \rho \rfloor - \lfloor \rho \rfloor \leq s\alpha + 1$ and, by the closed form of $d_s = \sum_{i < s} w_i 2^i 3^{\#\{\text{ones in } (i,s)\}}$ [3], each term at most $2^i \cdot 3^{(s-i-1)\alpha+1} \leq 3^{i\alpha} 3^{(s-i-1)\alpha+1} \leq 3 \cdot 3^{s\alpha}$ — the step $2^i \leq 3^{i\alpha}$ is exactly supercriticality. So (`dcoef_le_of_itin`, `tail_le_of_itin`)

$$d_s \leq 3s \cdot 3^{s\alpha}, \quad 2^s T^s(N) \leq 3^{s\alpha} (3N + 3s).$$

7.4 The level theorem

Theorem 7.2 (`sturmian_level`). *Let $2 \leq 3^\alpha$, $q\alpha = p + \delta$ with $0 < \delta < 1$, $Q \geq 3$, and suppose*

$$(L) \quad \|m\alpha\| \geq \delta \text{ for all } 0 < m < Q;$$

(T) every window of G consecutive integers contains a visit;

(S) for every $s \leq G$: $2^{Q-3+s} > 3^{(q+s)\alpha+1} (3N + 3s + 1)$.

Then N does not have itinerary w .

Proof. Let $n_0 < G$ be the first visit, $s = n_0 + 1$, $m = T^s(N)$. By (i) and (ii) the itinerary of m is q -periodic on its first $q+Q-2$ letters. The orbit of m is not q -periodic: otherwise w would be q -periodic from s on, so by (i) the visit indicator would be constant from s on, contradicting (T) (there is a visit) together with (ii) (two adjacent visits are impossible as $Q \geq 3$). Proposition 7.1 gives $2^{q+Q-2} \leq 2^q m + 3^o(m + 2^q)$ with $o \leq q\alpha + 1$, hence $2^{q+Q-2} \leq 2^{q+1} 3^{q\alpha+1} (m + 1)$. Multiplying by 2^s and using the tail bound together with $2^s \leq 3^{s\alpha}$,

$$2^{q+1} 2^{Q-3+s} \leq 2^{q+1} 3^{q\alpha+1} 3^{s\alpha} (3N + 3s + 1),$$

contradicting (S). □

Corollary 7.3 (Sturmian rung, conditional on Lagrange and three-distance). *Let $\alpha \in (\log 2 / \log 3, 1)$ be irrational, $\eta = \alpha \log_2 3 - 1 \in (0, 0.585)$, and suppose that for infinitely many positive k*

$$a_{k+1} > \frac{1 + 2\eta}{1 - \eta}.$$

Then no natural number has a Sturmian itinerary of slope α , with any intercept.

Proof. Apply Theorem 7.2 at such a level with $q = q_k$, $Q = q_{k+1}$, $G = q_k + q_{k+1}$; (L) is Lagrange's property, (T) is the three-distance theorem. For (S), take logarithms: since the coefficient $\alpha \log_2 3 = 1 + \eta$ of s on the right exceeds 1, the worst case is $s = G$, and

$$(Q-3+s) - ((q+s)\alpha+1) \log_2 3 - \log_2 (3N+3s+1) \geq q_{k+1}(1-\eta) - q_k(1+2\eta) - \log_2 (3N+3(q_k+q_{k+1})+1) - 5,$$

which, as $q_{k+1} \geq a_{k+1}q_k$, is at least $q_k(a_{k+1}(1-\eta) - (1+2\eta)) - O(\log q_k + \log N)$ and tends to $+\infty$ along the chosen levels. □

Concretely, partial quotients ≥ 2 infinitely often suffice for $\alpha < 0.789$, ≥ 3 for $\alpha < 0.883$, ≥ 4 for $\alpha < 0.946$, and ≥ 6 for every $\alpha < 1$ — a set of slopes of full Lebesgue measure in each range. Only the sign restriction (the level must be positive) and the golden-type slopes, whose partial quotients are eventually 1, are outside the corollary. Numerically (`analysis/sturmian_prefix_power.py`) the golden-type slopes are excluded just as well up to $\alpha \approx 0.9$: the returns to the visit interval take the two values q_{k+1} and $q_{k+1} + q_k$, the long return gives a run of length $\geq q_{k+1} + q_k - 2$, and the margin per q_k is then $\approx (a_{k+1} + 1)(1 - \eta) - (1 + 2\eta) - \eta$ (position of the first long return)/ q_k , which stays positive for η up to about 0.4. Proving that a long return occurs early enough is a refinement of the three-distance theorem that we have not carried out.

7.5 Scope and honesty

Theorem 7.2 is machine-verified with (L), (T), (S) as hypotheses. Everything about orbits, itineraries and sizes is inside Lean; the two Diophantine inputs and the asymptotic passage from (S) to a condition on partial quotients are not. Both inputs are nineteenth- and mid-twentieth-century classics with elementary proofs, and formalizing them (Lagrange's property

is close to Mathlib’s existing continued-fraction library; the three-distance theorem is not in Mathlib) is the natural next formalization target. With them, Corollary 7.3 becomes a theorem of the repository, and the Sturmian part of rung two closes for the stated slopes. The morphic, non-Sturmian words of rung two remain for the Mahler route.

Reproducibility

`lean/Collatz/Rung1.lean` builds against the repository’s pinned Mathlib with `lake build Collatz.Rung1; #print axioms reports propext, Classical.choice, Quot.sound` for the theorems `no_shifted_sigWord_itinerary`, `no_sigWord_itinerary` and `core`. `lean/Collatz/PrefixPower.lean` and `lean/Collatz/Sturmian.lean` build the same way and report the same axioms. `analysis/pade_rung1.py` recomputes the Padé certificates for degrees 2, 3, 4, 6, checks the predicted 2-adic valuations of $E(\alpha_k)$ against the series to 1053 bits, and prints the height ledger.

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